

Patient Safety Across Disciplines: Radiation Oncology Incident Learning System

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On June 19, 2014, the American Society for Radiation Oncology (ASTRO) and the American Association of Physicists in Medicine (AAPM) launched the Radiation Oncology Incident Learning System (RO-ILS). This is a national error reporting system that is voluntary and confidential. However, this program goes beyond just error reporting: participants are meant to collect and analyze their data and then institute corrective actions so that true incident learning takes place and clinic safety is improved. The RO-ILS initiative was started as part of ASTRO's Target Safely campaign. This campaign was designed to increase quality and safety in the field of radiation oncology by supporting a national error reporting system, increasing quality and safety content at national meetings, strengthening practice accreditation, increasing compatibility between vendors of technology, educating patients about what to ask their radiation oncologists, and advocating for passage of legislation to require licensing standards for personnel performing radiation treatments.

As of February 2015, there are 70 sites participating in RO-ILS and more than 320 events have been entered into the system. In its current form, this system is available to radiation oncology clinics at no charge. To participate, the institution must sign a contract with Clarity Patient Safety Organization (Chicago, IL). Although there is a standard contract, institutions and their attorneys may request modifications of this contract until both parties are mutually satisfied. Staff enter in events, and the quality improvement team or designated manager of the department is responsible for reviewing these events and taking the most appropriate action. The goal of this system is to disseminate errors of note and best practices to the radiation oncology community, to further our knowledge about the nature of error in radiation oncology, and to provide guidance to vendors about what improvements in commercial products may enhance patient safety.

So why is an incident learning system needed? It is widely known that medical errors are common. Medical errors are presently estimated to be the third leading cause of death in hospitalized patients.¹ Estimates within the field are that incidents in radiation therapy happen at a rate of 0.7% per treatment,² or one near miss per patient treatment course.³ As such, it is clear that radiotherapy incidents happen in every clinic. ASTRO and AAPM made the conscious choice to develop not just an error reporting system, but an incident learning system. The term incident was chosen to encompass not only events that reach the patient, but also to include variations in workflow, near misses, or unsafe conditions.⁴ Institutions with ro-

bust reporting systems are also using this system for practice improvement suggestions.

RO-ILS offers several benefits over a departmental incident learning system. ASTRO and AAPM partnered with Clarity Patient Safety Organization as a strategic move that was designed to afford participants in this system all of the protections that a patient safety organization (PSO) affords. The Patient Safety and Quality Improvement Act of 2005⁵ allowed for the creation of PSOs. The Agency for Healthcare Research and Quality oversees the PSO program. The Patient Safety Act affords confidentiality and privilege protections to the information contained within these systems. This means that the information is protected from discovery in legal proceedings. It directly addresses fears that safety event reports could be used against practitioners in malpractice cases. This was tested and upheld in federal court in Tinal versus Norton Healthcare,⁶ a case in which a pharmacist sought to produce these documents for use in an employment discrimination case. Despite the fact that these documents were sought not for malpractice, but for employment discrimination, the federal protections were upheld and the documents remained protected and undiscoverable.

RO-ILS, a national system, has additional benefits besides federal protections. As a specialty-specific incident learning system, RO-ILS has been tailored to serve the radiation oncology community. Data elements were developed on the basis of published consensus guidelines from within the field⁴ and carefully crafted by a steering committee. The Radiation Oncology Health Advisory Council, a group of volunteer subject matter experts and practitioners, oversees the collection and analysis of data. This group is made up of three radiation oncologists, three physicists, an administrator, and a dosimetrist, who serve for a

Table 1. Data on Survival Decrement Associated With Protocol Deviations in Clinical Trials

Disease Site of Protocol	Rate of Major Protocol Deviations		Decrement in Survival in the Presence of Protocol Deviations	
	%	Range	HR for Death	P
Non-small-cell lung ⁷	31		13% decrement*	
Head and neck ⁸	25		1.99	< .001
Pancreas ⁹	48		1.336	.0077
Meta-analysis ¹⁰	32	8 to 71	1.74	< .001

Abbreviation: HR, hazard ratio.
* No HR given.

1-year or 2-year term. The Radiation Oncology Health Advisory Council is charged with interpreting the data entered into RO-ILS and creating quarterly reports to RO-ILS participants. As data accumulate, best practices will be shared from the knowledge that is collected within this system.

Reports from experiences with incident learning systems demonstrate that with time and continued use of these systems, staff engagement increases.³ Safety culture metrics improve with time, including the belief that the department openly discusses ways to improve safety, the sense that reports are being used for safety improvement, and the sense that changes are being evaluated for effectiveness.³ Importantly, the rate of severe incidents decreased with time,² suggesting that use of these systems translates into meaningful improvements in patient safety.

How do errors in radiation oncology affect the quality of care? Errors in radiation therapy planning or delivery, regardless of their origin or mechanisms, ultimately lead to deviations from the prescribed intent of radiation dose or target volume. Interestingly, data exist on radiotherapy protocol deviations and their connection to patient outcomes. These deviations involve variations in prescribed dose or target volume of treatment. Throughout multiple protocols, including a meta-analysis, a consistent association was seen between presence of protocol deviations and decreased survival (Table 1). This suggests that proper delivery of radiotherapy treatment can be critical to quality and patient outcomes.

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RO-ILS has practical benefits for individual practitioners and practices. Participation in a PSO and error reporting are part of Standard 7 in the practice accreditation standards from ASTRO.¹¹ Participation in RO-ILS or another PSO can satisfy this requirement. Incident reporting is also specifically referenced in the American College of Radiology's practice accreditation requirements.¹² Participation in RO-ILS is approved as a practice quality improvement project for maintenance of certification for the American Board of Radiology.¹³

RO-ILS is an integral part of any quality management program, with benefits in safety culture, decreases in the rate of severe incidents, and advancement of best practice dissemination within the field, and is housed in a confidential, protected environment.

Author's Disclosures of Potential Conflicts of Interest

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